

Implementation Documentation

Playback Speed and Mind-Wandering Experiment

This document describes the software implementation and the materials required to prepare, run, test, understand, and continue development of the experiment application. The separate README.md contains the minimum installation and demo instructions.

1. Implementation Overview

The implementation is a local browser-based application built with React and Vite. It manages the complete software-controlled experiment flow: participant setup, condition assignment, video playback speed, randomized mind-wandering prompts, comprehension quizzes, result storage, and CSV export.

2. Main Functionality

2.1 Counterbalanced Conditions

The application supports two experimental groups. Playback speed is controlled automatically by the software.

Group	Condition 1	Condition 2
Group 1	Video 2 at 1.5x	Video 1 at 1.0x
Group 2	Video 2 at 1.0x	Video 1 at 1.5x

2.2 Mind-Wandering Prompts

Each video is divided into four equal quarters. One prompt time is generated randomly within each quarter. A 15-second buffer is applied at both ends of each quarter so that a prompt does not appear immediately at a quarter boundary.

When a scheduled prompt time is reached, the video pauses and the participant is asked whether they were mind-wandering. If the participant selects No, the response is saved and playback resumes. If the participant selects Yes, a second question asks whether the thoughts were related or unrelated to the lecture. After that response is recorded, playback resumes automatically.

2.3 Comprehension Quizzes

Each video is followed by a five-question multiple-choice comprehension quiz. Each correct answer is worth one point, producing a score from 0 to 5. The application automatically selects the quiz associated with the video that has just finished. All five questions must be answered before submission. The score is recorded but is not shown to the participant.

2.4 CSV Export

After both video conditions and both quizzes are completed, the application provides a Download results button. The file is named using the participant ID, for example P01_results.csv.

3. Source Code Structure

The core project structure is:

```

| playback-experiment/
| ├── public/
| |   ├── videos/
| |   |   ├── video1.mp4
| |   |   └── video2.mp4
| |   └──
| ├── src/
| |   ├── App.jsx
| |   ├── App.css
| |   └── main.jsx
| ├── package.json
| ├── package-lock.json
| └── vite.config.js

```

3.1 App.jsx

App.jsx contains the main experiment logic, including participant ID entry, group selection, condition order, video selection, playback-speed assignment, prompt scheduling and triggering, mind-wandering responses, quiz selection and scoring, experiment progression, result storage, and CSV generation.

3.2 App.css

App.css contains the styling for the start screen, instruction screens, video view, prompt overlay, quiz interface, and completion screen.

3.3 main.jsx

main.jsx is the React application entry point.

4. Dependencies and Build Environment

The project uses npm for dependency management. All required JavaScript dependencies and their exact versions are defined in package.json and package-lock.json. No database, backend framework, external API, or cloud service is required.

Install all required dependencies with:

```
npm install
```

Run the development version with:

```
npm run dev
```

Create a production build with:

```
npm run build
```

Preview the production build with:

```
npm run preview
```

5. Latest Stable Version and Version Control

The submitted source code should represent the latest stable version used for the experiment. Because this is a browser-based React application, a separate .exe or .apk is not required. A deployable production version can be generated with npm run build and is written to the dist/ directory.

Github repository:

<https://github.com/jakobdue/COMPSCI705Project.git>

6. Data Output

The CSV contains three row types: prompt-timing, mind-wandering, and quiz.

Field	Description
type	Row type: prompt-timing, mind-wandering, or quiz
participantId	Participant identifier
group	Experimental group
conditionNumber	First or second condition
video	Video identifier
videoTitle	Full video title
speed	Playback speed
videoDuration	Video duration in seconds
prompt1Scheduled	Scheduled time of prompt 1
prompt2Scheduled	Scheduled time of prompt 2
prompt3Scheduled	Scheduled time of prompt 3
prompt4Scheduled	Scheduled time of prompt 4
promptNumber	Prompt number from 1 to 4
scheduledPromptTime	Randomized target time for the prompt
actualPromptTime	Video time when the prompt was actually displayed
mindWandering	true or false
related	related, unrelated, or blank
score	Quiz score from 0 to 5
q1Answer - q5Answer	Participant quiz responses stored as A, B, C, or D

7. Design Decisions

7.1 Local Video Playback

The videos are stored locally rather than embedded from an external service. This reduces variation caused by internet connectivity, buffering, advertisements, recommendations, and external interface changes.

7.2 Local CSV Export

A database was not required for the small, supervised study. Data is temporarily stored in React state and exported to a local CSV file when the experiment finishes.

7.3 Standard Video Controls

The standard browser video controls remain enabled. Participants are instructed not to skip forward or backward or change the playback speed. The implementation is intended for supervised use.

8. Known Limitations

- Video seeking is not technically disabled; compliance relies on participant instructions and supervision.
- Current session data is stored in browser memory until export. Refreshing or closing the page before download can lose the session data.

- The application does not automatically upload results to a central database.
- The application is intended to run locally and has not been configured as a publicly hosted production service.

9. Troubleshooting

npm is not recognised: Confirm that Node.js is installed. Run `node -v` and `npm -v`. On Windows PowerShell, script execution settings may block `npm.ps1`.

Video shows 0:00 and does not play: Confirm that the files exist exactly at `public/videos/video1.mp4` and `public/videos/video2.mp4`. Folder-name casing can matter in some environments.

Video and quiz do not match: Confirm that `video1.mp4` is the Robert Waldinger talk and `video2.mp4` is the Karen Faith talk.

No CSV is downloaded: The download option appears only after both video conditions and both quizzes are completed.

Session data disappears: The current implementation stores data in browser memory. Refreshing or closing the page clears the current session.

10. Authors

COMPSCI 705 Group Project

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